Jitao Zhou

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Research interests: AI for corrosion science, physics-constrained alloy inverse design, materials knowledge systems, multi-agent collaboration and medical image analysis.

EDUCATION

University of Emergency Management

Undergraduate studies in Material Forming and Control Engineering Sep. 2022 – Jun. 2026

- Transcript institution: North China Institute of Science and Technology; School of Emergency Equipment.
- Official CGPA: 3.03/5.00; credit-weighted average: approximately 80.30/100, calculated from graded courses on the transcript.
- Relevant coursework: Fundamentals of Materials Science, Principles of Material Forming, Material Forming Inspection and Control, Finite Element Analysis, Fundamentals of Nondestructive Testing, and C Programming.

PUBLISHED ARTICLES

Pan Li, **Jitao Zhou**, Huihui Sun, Ying Wei, Qiqi Yang (2026). Thermodynamics-constrained generative adversarial network for synergistic strength-ductility inverse design of rare-earth magnesium alloys. *Materials Today Communications*, 52, 115076. DOI: 10.1016/j.mtcomm.2026.115076.

Ruoyu Wu, Jing Shi, **Jitao Zhou*** (2026). Kidney Tumor Segmentation With a Multistage Adaptive Boundary-Aware Network. *Annals of the New York Academy of Sciences*, 1557(1), e70231. DOI: 10.1111/nyas.70231.

Pan Li, **Jitao Zhou**, Huihui Sun, Penglin Li, Xi Chen (2026). GBDR-Net: A YOLOv10-Derived Lightweight Model with Multi-Scale Feature Fusion for Accurate, Real-Time Detection of Grape Berry Diseases. *Horticulturae*, 12(1), 38. DOI: 10.3390/horticulturae12010038.

Pan Li, **Jitao Zhou***, Jian Zeng, Qian Zhao, Qiqi Yang (2025). MSWindD-YOLO: A Lightweight Edge-Deployable Network for Real-Time Wind Turbine Blade Damage Detection in Sustainable Energy Operations. *Sustainability*, 17(19), 8925. DOI: 10.3390/su17198925.

Pan Li, **Jitao Zhou**, Huihui Sun, Jian Zeng (2025). RDRM-YOLO: A High-Accuracy and Lightweight Rice Disease Detection Model for Complex Field Environments Based on Improved YOLOv5. *Agriculture*, 15(5), 479. DOI: 10.3390/agriculture15050479.

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MANUSCRIPTS UNDER REVIEW AND REVISION

Jitao Zhou, Zihui Zhang, Dawei Zhang, Ruoyu Wu, Liang Qiao*. UFB-PolypNet: Disagreement-Aware SAM Distillation for Lightweight Cross-Dataset Polyp Segmentation. *Annals of the New York Academy of Sciences*. Major Revision; first author.

Pan Li, **Jitao Zhou***, Xiaowei Fu, Penglin Li. AeroTrustNet: An EASA-Roadmap-Informed Framework for Trustworthy Aircraft Surface Defect Detection. *Annals of the New York Academy of Sciences*. Major Revision; corresponding author. Manuscript ID: 7026592.

Haiyan Gong, **Jitao Zhou***, Hao Tang, Yageng Li, Yuxin Tang. GeoTHA-YOLO: An Efficient Geometry-Aware Network for Pelvic Keypoint Detection in Clinical Assessments Following Total Hip Arthroplasty. *Biomedical Signal Processing and Control*. Under Review; corresponding author. BSPC-D-26-14915.

* Corresponding author; manuscript statuses updated on 2 October 2026.

RESEARCH EXPERIENCE

University of Science and Technology Beijing **Beijing, China**

Research Assistant (RA), AI and Materials Research May 2026 – Present

Materials 2030 Major Project Tasks

Project lead: Prof. Dawei Zhang.

- Responsible for agent and AI-ready data modules within a materials-in-service platform, connecting multisource data preparation, structured representation and model access.
- Building a RAG Hub platform for knowledge retrieval from materials literature and research data to support materials-in-service analysis.
- Developing a multi-agent collaboration system with task orchestration and module integration for knowledge retrieval, data processing and model analysis.

National Key R&D Program: Data-Driven Artificial Joint Materials and Components

Project No. 2024YFB3817500; project lead: Yageng Li.

Inverse Composition Design of CoCrMo Alloys

- Undertaking CoCrMo inverse composition design within Haiyan Gong's assigned tasks, combining physics-informed neural networks (PINNs) and GANs to generate candidate compositions.
- Using Thermo-Calc phase-diagram calculations for theoretical validation; the manuscript is complete and experimental validation is underway.

Beijing Jishuitan Hospital Collaboration: Post-THA Imaging

- Organized and completed GeoTHA-YOLO experiments using 271 postoperative pelvic radiographs provided by Beijing Jishuitan Hospital with five-fold cross-validation.
- Combined anatomical geometric priors, lightweight networks and large-model knowledge distillation for keypoint detection and automated calculation of post-THA clinical parameters.

Dynamic Atmospheric Corrosion and Corrosion-Resistance Control of Low-Alloy Steels

National Natural Science Foundation of China Young Scientists Fund (Category C); project lead: Haiyan Gong.

- Integrating macroscopic surface morphology with microscopic corrosion-product features to develop a long-sequence, multi-model forecasting framework for corrosion-state and morphology evolution over specified time horizons.
- Exploring continuous-state corrosion modelling with a world-model-inspired evolution forecasting approach to support dynamic mechanism analysis and corrosion-resistance control.

RESEARCH EXPERIENCE CONTINUED

Robust Object Detection in Complex Environments

Provincial research project QN2025015 Apr. 2023 – Present

- Designed spatial attention, multiscale feature fusion and backbone modifications for visual detection in oil-well inspection and disaster search-and-rescue scenarios.
- Developed context-enhanced and noise-resistant feature representations for uneven illumination, imaging interference and feature drift; optimized robustness and inference cost.
- Adapted detection models to embedded sensing platforms, developed acceleration interfaces, and coordinated validation under operating conditions with the COSL oilfield technology manufacturing centre.

Airborne Detection of Wind Turbine Blade Damage

Department-level research project 2024013009 Sep. 2024 – Present

- Introduced and tuned Shape-IoU loss for elongated and irregular damage to improve localization and training convergence.
- Conducted INT8/FP16 deployment experiments on NVIDIA Jetson Nano/NX, integrated sensing, detection and feedback, and participated in wind-farm validation.

Luoyang Institute of Special Materials, Shanghai Jiao Tong University

Research Assistant Jun. – Aug. 2024

- Contributed to drafting and submitting 10 invention patent applications concerning rare-earth magnesium alloy spinning processes, equipment and industrial automation.
- Collaborated with materials researchers to apply machine learning to manufacturing quality control and spinning process parameter optimization.

PROFESSIONAL EXPERIENCE

Nashijia Automation Technology (Wuxi) Co., Ltd.

Algorithm Technology Consultant May 2025 – Present

- Contributed to Halcon-based industrial defect inspection and Faster R-CNN adaptation; developed point-cloud preprocessing, registration and surface reconstruction workflows.
- Led development of an LLM-driven CAD platform using FreeCAD and MCP, translating natural-language instructions into API calls for parametric modelling.

WorldQuant

Alpha Research Consultant Aug. 2025 – Present

- Deployed local LLMs with Ollama and developed workflows for alpha generation, validation and screening; developed and open-sourced agent tools for factor generation and backtest analysis.

ADDITIONAL PROJECTS AND TEACHING SUPPORT

Collaborative Training for Emergency Equipment Engineering

Department-level project GJXHZ2024-30 Sep. 2024 – Present

- Contributed to industry needs analysis, competency modelling and collaborative training design; helped translate visual inspection research into three practical course modules.

Reform of Undergraduate Thesis Training

Provincial teaching reform project 2022GJJG478 Jan. 2023 – Dec. 2025

- Contributed to training design integrating engineering ethics and practice, development of an engineering case library, and thesis quality evaluation.

Competency Development in Emergency Equipment Engineering

University project HKJYGH202307 Jul. 2023 – Jun. 2024

- Contributed to competency indicators and curriculum integration; developed three comprehensive laboratory guides based on visual detection and robot control research.

Robotics Technology and Applications Course Development

University course development project HKJG202410 Jul. 2024 – Present

- Assisted the lead instructor with course planning, integrating robot kinematics and dynamics with engineering ethics.

CERTIFICATES AND AWARDS

- National Scholarship:2022-2023.
- Second Prize, 16th International Advanced Robotics and Simulation Technology Competition.
- Third Prize, AI and Robot Creative Design category, Hebei regional round of the 2023 North China Five-Province University Student Robotics Competition.
- Second Prize, Hebei regional round of the 14th MathorCup Mathematical Application Challenge, 2024.

SKILLS AND LANGUAGES

- Programming and computing: Python, PyTorch; familiarity with C++; MATLAB/Simulink.
- Computer vision: YOLOv5/v10, Faster R-CNN, image segmentation, attention, multiscale fusion, Shape-IoU and lightweight architectures.
- AI for materials design: PINNs, GANs, inverse alloy composition design and Thermo-Calc-assisted phase-diagram validation.
- LLMs and automation: RAG Hub, AI-ready data, Ollama, prompt engineering, MCP, multi-agent collaboration and task orchestration.
- Industrial vision and deployment: Halcon, point-cloud registration and surface reconstruction, NVIDIA Jetson Nano/NX, INT8/FP16 quantization.
- Engineering modelling: SolidWorks, AutoCAD, FreeCAD and API-driven modelling; mechanical design, kinematic and dynamic simulation.
- Languages: Chinese (native); English (CET-6, 461).